

# Bora Hwang

## Agri\_nova analysis1

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## AGRI NOVA MARKET ENTRY ANALYSIS

### An Econometric Study of Revenue Determinants and Product Launch Strategy

**Name:** Abdullah Ibrahim

**Course:** Economics And Business Analytics

**Assignment:** Individual Report (40%)

**Date:** 21 April 2026

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#### 1. Management Summary

This study looks at the primary economic variables that affect sales income for CropSense and FreshConnect, two products that AgriNova has suggested. Regression analysis is used to determine which product should be introduced first based on data-driven insights and to identify what generates revenue for each product. With extremely high adjusted R<sup>2</sup> values, the results show that both models successfully explain revenue. But the factors that affect each product are very different.

Prices for agricultural commodities and advertising have a major impact on CropSense's revenue. Another important factor is price, therefore changes in price have a significant impact on demand. This suggests that pricing choices must be handled carefully. Further scenario analysis reveals that raising digital advertising and changing prices might increase income by about 6.6%.

On the other hand, consumer-related variables like household income, digital advertising, and population density have a greater impact on FreshConnect. One important discovery is that FreshConnect's demand is not highly sensitive to price fluctuations, making price tweaks easier to implement without significantly affecting demand.

All things considered, FreshConnect seems to be the better choice for the first launch. It has less short-term risk, depends on steady demand drivers, and provides more steady and predictable revenue.

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#### 2. Introduction (Industry Overview)

Both the fresh food delivery market and the agritech industry have grown significantly in recent years. Urbanization, shifting consumer demands, and technology developments have been the main drivers of this expansion.

Farmers are becoming more and more interested in agritech solutions, such as IoT-based monitoring systems and smart agricultural tools. These technologies increase productivity,

cut waste, and maximize the use of resources like fertilizer and water. Concurrently, urban consumers

desire convenient delivery of fresh, high-quality meals. The idea of "farm-to-table" has gained popularity, particularly in urban areas where consumers place a high value on convenience and food sourcing transparency.

Southeast Asian governments have been instrumental in fostering these advancements. They have promoted the adoption of contemporary farming technologies through infrastructural investments, digital agriculture initiatives, and subsidies.

The two marketplaces have different demand patterns from an economic perspective. Due to financial constraints, farmers, who are the main users of agritech products like CropSense, are typically more price-sensitive. On the other hand, urban consumers who purchase fresh food services like FreshConnect are frequently more concerned with convenience and quality than with price.

Making business decisions requires an understanding of these distinctions. According to economic theory, the law of demand states that demand often declines as prices rise. However, depending on the product and target market, price elasticity—the degree to which demand reacts to price changes—varies greatly.

One useful method for analyzing these interactions is regression analysis. It enables companies to evaluate how different elements, such as income, advertising, and price, affect revenue. Businesses can improve their pricing, marketing, and product planning decisions by doing this.

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### 3. Report Objectives

This report's primary objective is to determine and assess the economic aspects influencing CropSense and FreshConnect's sales revenue.

More precisely, the paper seeks to:

- Determine which important factors have a major influence on revenue.
- Estimate the price elasticity of demand for both products
- Analyze the effects of digital and traditional advertising
- Evaluate the regression models' performance using  $R^2$  and adjusted  $R^2$
- Conduct scenario analysis to explore potential revenue changes
- Recommend the most suitable product for initial launch



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## 4. Data Analysis

### 4.1 Exploratory Analysis and Bivariate Relationships

The first stage of the research was to use simple linear regression to look at each variable's relationship to revenue.

With an  $R^2$  of about 0.75, the commodity price index explains the greatest portion of revenue volatility for CropSense. This indicates a high correlation between the performance of the product and the state of the agricultural market. While price and wealth seem to have less individual effects, advertising variables—both digital and traditional—have a moderate impact.

FreshConnect, on the other hand, shows an alternative pattern. With an  $R^2$  of almost 0.98, household income and revenue have a very significant correlation. With an  $R^2$  of about 0.91, digital advertising also has a big impact. Price appears to have virtually no explanatory power on its own, and population density has a smaller impact.

These findings imply that FreshConnect is largely influenced by consumer-related factors, while CropSense depends more on external market conditions.

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### 4.2 Multiple Regression Model Results

The outcomes of a multiple regression model become more understandable when all variables are included.

With an adjusted  $R^2$  of roughly 0.99 for CropSense, the model explains almost all of the variation in revenue. According to economic theory, price has a statistically significant negative impact. The commodity price index continues to be the most significant factor, but both traditional and digital advertising have significant beneficial benefits. It's interesting to note that in this model, household income is not statistically significant.

With an improved  $R^2$  near 0.999, the model performs even better for FreshConnect. Once more, price has a detrimental impact, albeit a minor one. Demand is largely driven by household income, and digital advertising has a highly favorable effect. Another important factor is population density, which suggests that the product works especially well in cities.

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### 4.3 Price Elasticity Analysis

Demand sensitivity to price fluctuations is measured by price elasticity.

The elasticity for CropSense is roughly -0.39. This implies that demand is rather elastic, which means that changes in price will have a discernible effect on revenue and demand.

The elasticity for FreshConnect is significantly smaller, at about -0.02. Customers are not extremely responsive to price fluctuations, which suggests that demand is highly inelastic. This distinction has important ramifications. While FreshConnect offers more flexibility, CropSense necessitates cautious price judgments.

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#### 4.4 Statistical Significance and Confidence Intervals

At the 5% level, the majority of the variables in both models are statistically significant. This indicates that the observed associations are not likely to be the result of chance.

Confidence intervals support these conclusions even more. They attest to the fact that price has a negative influence on revenue whereas advertising has a significant favorable impact.

Income and population density also have significant and consistent influence on FreshConnect. All things considered, the models are statistically sound and offer trustworthy insights.

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#### 4.5 Economic Logic Validation

The analysis's findings are consistent with fundamental economic theory.

The law of demand is reflected in the inverse relationship between price and demand. The beneficial effects of advertising show how marketing can increase demand and customer awareness. Demand for FreshConnect is favorably influenced by income, suggesting that it is a typical good. Finally, the importance of commodity pricing for CropSense shows how demand can be influenced by market-specific factors.

The models are more credible since they are consistent with theory.

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#### 4.6 Scenario Analysis

Two scenarios were tried for CropSense in order to gain a better understanding of how changes in important variables impact revenue.

In the first case, revenue increased by roughly 6.64% as a result of lower prices and more digital advertising. In the second case, a lesser revenue boost of 0.51% resulted from cutting marketing activities while little altering economic conditions.

These findings demonstrate that digital marketing and pricing are essential instruments for raising performance.

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## 5. Discussion

It is evident from the analysis that CropSense and FreshConnect function rather differently.

CropSense is more reliant on outside variables, especially the cost of agricultural commodities. Because of its modest price elasticity, pricing choices are important and have a big influence on demand. Higher gains are possible as a result, but there is also more risk.

In contrast, FreshConnect is more reliable and consistent. Income, advertising, and population density—all of which are comparatively constant—are what propel its demand. Pricing adjustments have little impact on demand due to the low price elasticity, which facilitates management.

FreshConnect is the safer option from a business standpoint. It provides scalability and consistent growth, particularly in metropolitan markets. CropSense has the potential to be profitable, but it needs more careful planning and is more susceptible to outside influences.

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## 6. Recommendations

Numerous suggestions can be made in light of the results.

First, the first product to be released should be FreshConnect. It is a less risky choice due to its steady demand and strong predictability.

Second, each product should have a different price strategy. While CropSense requires a more cautious approach, FreshConnect permits flexible pricing.

Third, since digital advertising has a significant impact on both products' revenue, the business should give it top priority. Nonetheless, CropSense should continue to use traditional advertising.

Fourth, it's important to define target markets precisely. CropSense should target agricultural areas, whereas FreshConnect should concentrate on affluent urban areas.

Finally, if more data is available and market conditions are better understood, CropSense may be introduced.

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## 7. Conclusion

Regression analysis was performed in this study to assess the key variables affecting CropSense and FreshConnect revenue. Even if both models did well, the outcomes nevertheless show how each product differs and has opportunities.

CropSense necessitates careful price selections and is impacted by the state of the agricultural market. In contrast, FreshConnect exhibits steady, predictable demand and is driven by consumer factors.

The data suggests that FreshConnect is a superior choice for an initial launch. It provides higher growth potential, more flexibility, and reduced risk.

All things considered, data-driven research offers insightful information that can assist AgriNova in making more informed strategic choices.

## 8. References

- Basic Microeconomic Theory (Demand and Elasticity)
- Regression Analysis Methods
- Course Lecture Notes

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## 9. Annexures

- $R^2$  Comparison Table
- Regression Outputs
- Confidence Intervals
- Scenario Analysis Results